Project Report:

Apple Leaf Disease Classification using Random Forest

1. Objective:

The purpose of this project is to classify apple leaf images into four categories:

- Apple Apple Scab
- Apple Black Rot
- Apple Cedar Apple Rust
- Apple Healthy

Using traditional machine learning techniques, the goal is to build a reliable image classification model based on pixel data extracted from the images. The project focuses on applying the Random Forest algorithm, combined with hyperparameter tuning, to achieve high accuracy without relying on deep learning.

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2. Dataset Description:

The dataset consists of images organized into four folders, each corresponding to one of the above classes. Each image is a coloured photo of an apple leaf showing disease symptoms or healthy conditions. The folders are stored in a master folder titled "Image Data base".

Each image is used as a sample in the dataset, and its folder name serves as the label.

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3. Methodology:

3.1. Image Preprocessing:

To ensure consistency and reduce computational load:

- All images were resized to 64×64 pixels.
- The colour format was converted from BGR (OpenCV default) to RGB.
- Each image is flattened into a 1D array of 12,288 features.

3.2. Label Encoding:

Each folder name was treated as a class label and encoded numerically using a label encoder for compatibility with machine learning algorithms.

3.3. Data Splitting:

The dataset was split into:

- 80% for training
- 20% for testing

The split was stratified to ensure class balance in both subsets.

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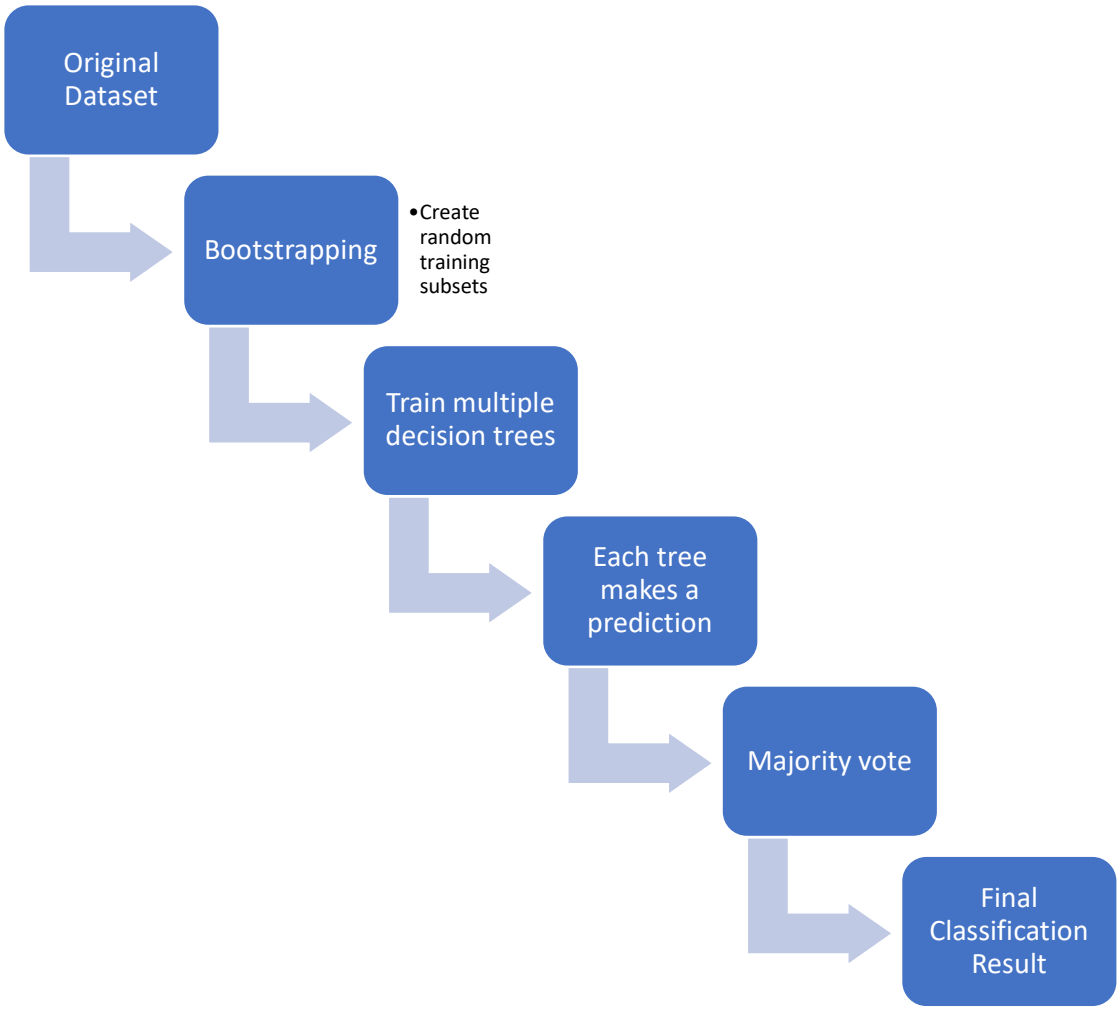
4. Random Forest Algorithm:

Random Forest is an ensemble learning technique that builds multiple decision trees and combines their results to improve prediction accuracy and prevent overfitting.



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5. Hyperparameter Tuning Using GridSearchCV:

GridSearchCV is used with 5-fold cross-validation to tune the following parameters:

- n_estimators: [50, 100]
- max_depth: [None, 10]

Best Parameters:

{‘max_depth: 10, ‘n_estimators’: 100}

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6. Model Evaluation:

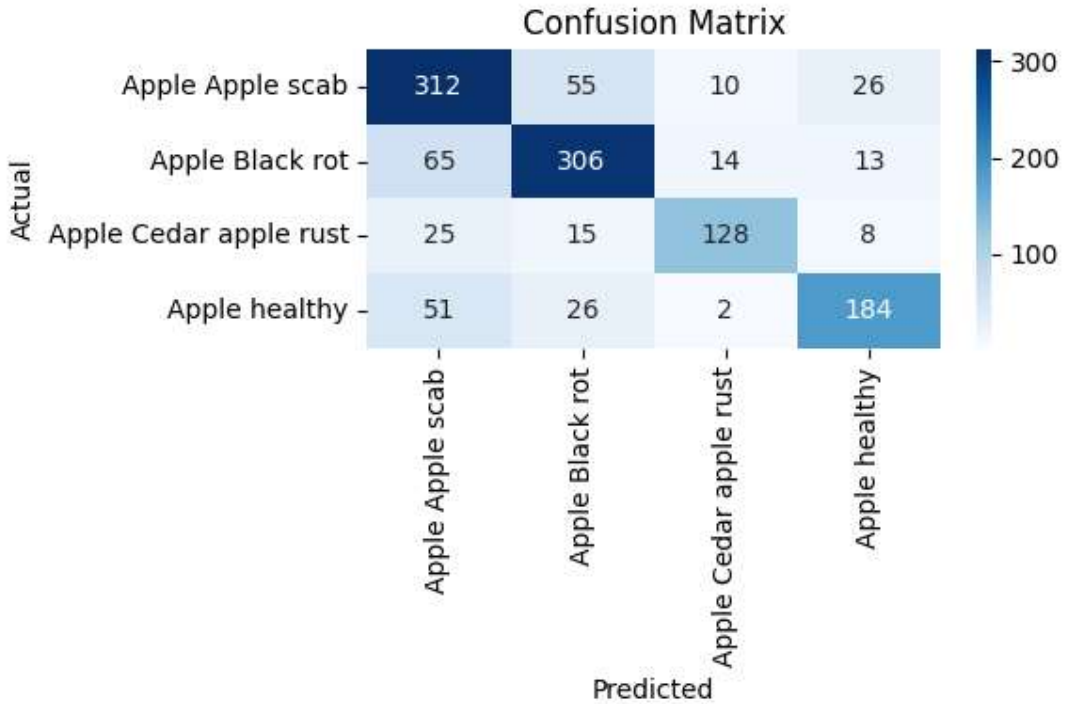
The final model was evaluated using the test dataset.

6.1. Accuracy:

The model achieved a test accuracy of 75%, indicating good generalization on unseen data.

6.2. Confusion Matrix:

The confusion matrix provided a visual representation of how well the model performed across each class. Most classes were classified correctly, with a few minor misclassifications between visually similar diseases.



6.3. Classification Report:

Precision, recall, and F1-score were calculated for each class. The scores showed balanced and consistent performance across all categories.

Class	Precision	Recall	F1-Score	Support
Apple Apple Scab	0.69	0.77	0.73	403
Apple Black Rot	0.76	0.77	0.77	398
Apple Cedar Apple Rust	0.83	0.73	0.78	176
Apple Healthy	0.80	0.70	0.74	263
Weighted Average	0.76	0.75	0.75	1240

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7. Prediction on New Images

A separate image prediction function was implemented to allow users to input any new apple leaf image and receive an output label indicating its disease category. The image undergoes the same preprocessing as the training data before being fed into the model for classification.

The model correctly predicted disease categories for test images, validating its usability in real-world applications.

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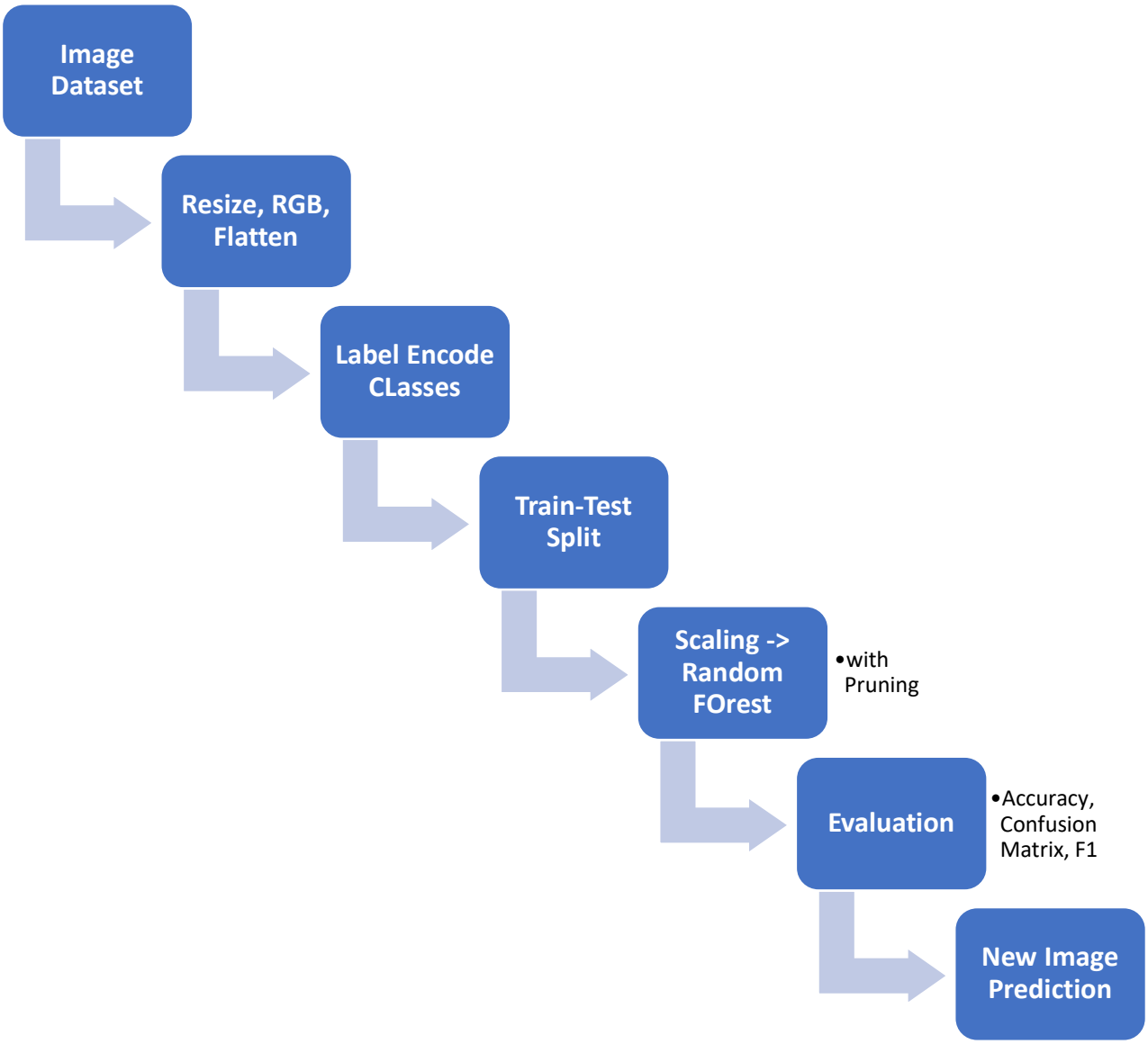
8. Project Workflow Summary



9. Conclusion

The Random Forest model, with tuned parameters (max_depth=10, n_estimators=100), achieved a **75% test accuracy**. Precision, recall, and F1-scores were balanced across all four classes, with particularly strong performance on **Apple Cedar Apple Rust**. The model is capable of generalizing well to unseen data and is suitable for use in a leaf disease classification pipeline.